Final Project Design Phase

Computer Science - Week 11

Oct 3, 2026 - Version 0.0.4

For complete instructions and project ideas, refer to the assignment in Canvas. Specifically, review the final project requirements before you continue below. If you have not reviewed the project requirements in Canvas (you might have to go to the modules view and check the current or past week's content), you will not be able to properly complete the parts below. In addition, it may be helpful to review the information on this Drafter documentation page.

0) Project Planning and Preparation

Successful programmers spend time planning before they begin coding. *After* reviewing the project requirements, answer the following questions to help guide your design and development process.

A. What concepts from class or previous assignments might help you complete this project? Which programming concepts (loops, conditionals, lists, classes, etc.) do you expect to use? What programming skills do you want to improve while completing this project?

I expect to use programming concepts such as conditionals and dataclasses. I want to improve my ability to design webpages and learn how to upload images to a webpage.

B. Which parts of the project do you expect to be the easiest for you? The hardest? Why?

I expect that writing out the procedure in general will be the easiest because I can mostly copy from a lab manual. I expect that the hardest thing will be to figure out how to modify the procedures based on user input because that will include understanding the procedures well enough to calculate the necessary quantities and to change the procedures based on the specific goal.

C. Even though you have not yet designed the final project, estimate which tasks will likely require the most time and effort? Which features or components should be implemented first?

I expect that the most time and effort will be devoted to figuring out the calculations for the different quantities and figuring out how to upload images to the website (as well as making the images in BioRender). I think I will first implement the basic page structure and the flow of selecting procedures/desired outputs and quantities.

D. What distractions might interfere with your work and how will you minimize them? What resources will you need available while working? What questions, if any, do you still have about the assignment requirements?

All of my other classes and also trying to code while sitting in a room with my friends will all distract me from my work. To minimize this, I will have to isolate myself in a room (with Jhayce, of course, so I can ask him questions) for a set period of time to code. Jhayce will be one of my resources, but I will also need access to all the previous CS assignments, the project description, the Drafter documentation website, and the internet. I have many questions about the assignment requirements that I can't formulate and that would probably be solved if I actually read the assignment requirements.

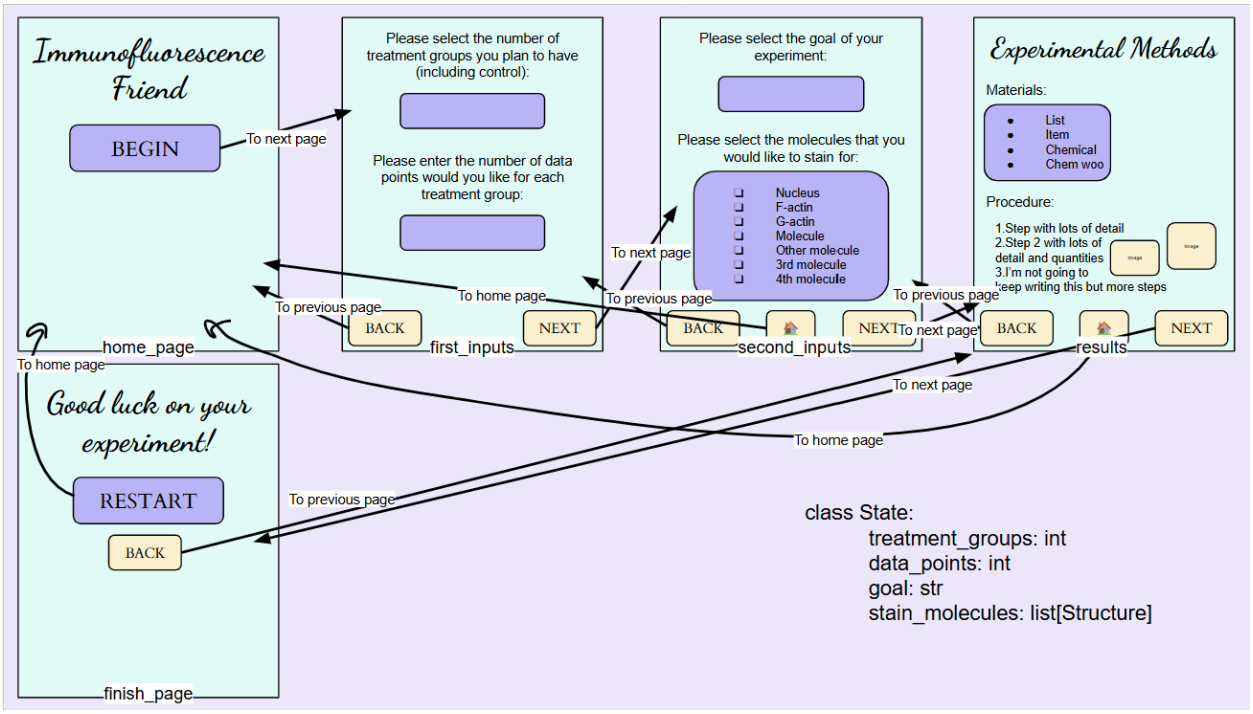
1) Site Diagram

Create a mock-up of your planned website. We suggest you make the mockup on paper, although you can also use software like PowerPoint or Slides to make a mockup. There is other design software that could also be used, but we strongly encourage you to keep it simple. Do not spend more than 15-30 minutes detailing the mock-up. Your goal should be to make a very quick and dirty diagram, to show that you have some kind of idea.

We recommend drawing each page as a rectangle, and laying out the text, buttons, and other components inside. Then, use arrows to indicate links between pages. Label each page the way you will name each function, and then put labels on the arrows to indicate what the button links. Also write down any parameters that should come into the pages.

In addition to the pages and routes, write down what fields your `State` will have, both their names and types. Think clearly about what data will need to be persistent across the entire website.

Add photo(s) or screenshot(s) of your website mock-up and labeled diagram on the next page.


Immunofluorescence Friend
BEGIN
To next page
home_page
To home page
Good luck on your experiment!
RESTART
BACK
finish_page
Please select the number of treatment groups you plan to have (including control):
Please enter the number of data points would you like for each treatment group:
To previous page
BACK
NEXT
first_inputs
To home page
To next page
Please select the goal of your experiment:
Please select the molecules that you would like to stain for:
Nucleus
F-actin
G-actin
Molecule
Other molecule
3rd molecule
4th molecule
To previous page
BACK
NEXT
second_inputs
To next page
Experimental Methods
Materials:
List
Item
Chemical
Chem woo
Procedure:
1.Step with lots of detail
2.Step 2 with lots of detail and quantities
3.I'm not going to keep writing this but more steps
To previous page
BACK
NEXT
results
To next page
To home page
To previous page
class State:
treatment_groups: int
data_points: int
goal: str
stain_molecules: list[Structure]

2) Implementing Requirements

This table should serve as a draft for how you will implement the requirements into your final project. You may change your ideas later on, but your final site must fulfill all the requirements.

Requirement	**Implementation**
Purpose/Functionality A 1-2 sentence description of the purpose or functionality of your site	The purpose of this website is to be an interactive and instructive lab manual for the lab procedure of immunofluorescence staining followed by confocal microscopy. The website is intended for novice researchers to be able to better understand the procedure they intend to perform and have easy access to clear instructions, illustrations, and calculations of the quantities they need.
State Dataclass Attributes Minimum of 4 attributes, one of which must be a list/dict of instances of your second dataclass. Briefly describe each attribute, its type, and how it will be meaningfully modified throughout the site.	class State: treatment_groups: int data_points: int goal: str stain_molecules: list[Structure] treatment_groups is a user input of the number of different treatment groups they will have in their experiment, including the control. The str input will be turned into an int and modified mathematically for the material quantity calculations. data_points is a user input of the number of data points they want to have for each treatment group. The str input will be turned into an int and modified mathematically for the material quantity calculations. goal is a user input for the goal of the experiment, which will be selected from a dropdown menu. This could be something like “Compare ratio of G-actin to F-actin”, and it will be used to modify the procedure on page 4. stain_molecules is a list of all of the molecules that the user wants to stain for, selected from a list of checkboxes. Each molecule will have a dataclass instance including information about its staining molecule, which will be used for both the material

	quantity calculations and to modify the procedure on page 4.
Second Dataclass Your second dataclass, including its attributes, their types, and how they will be meaningfully modified (if they are modified)	class Structure: structure_name: str staining_molecule: str ratio: int structure_name identifies the name of the structure that is being stained for. It is not meaningfully modified but informs the wording of the procedure. staining_molecule identifies the molecule that will be used to stain for the identified structure. It is not meaningfully modified but informs the wording of the procedure. ratio is a number that is mathematically modified to calculate the quantities of the staining molecule used. The staining molecule will be mixed with Permblock at a 1:ratio ratio.
Routes (minimum 7) The names of your route functions	start next_to_second_inputs next_to_results next_to_finish home back_to_first_inputs back_to_second_inputs back_to_results
Pages (minimum 5) A brief description of the pages in your site	home_page: displays title, welcome message, and button to begin first_inputs: asks for user inputs (text boxes) regarding number of treatment groups and number of data points desired second_inputs: asks for user inputs regarding the goal of the experiment (from a dropdown menu) and the molecules they would like to stain for (from checkboxes) results: displays the materials/quantities needed, a detailed procedure using the molecules they

	selected, and images showing parts of the procedure finish_page: displays a hopeful end message, a button to restart, and a button to go back to the procedure page
User Input (minimum 3) Description of user input components (TextBox, TextArea, CheckBox, or SelectBox) and their purposes For more information on these input types, see Drafter Documentation.	TextBox: the user inputs a str representing an int representing the number of treatment groups they want in their experiment TextBox: the user inputs a str representing an int representing the number of data points they want to end up with per treatment group SelectBox: the user selects from a list of 3-5 options what the goal of their experiment is CheckBox: the user selects from a list of checkboxes which molecules/structures they would like to stain for (perhaps with recommendations coming from their goal selection)
If Statements (minimum 3) Brief description of what will be evaluated in your if statements and how they will meaningfully change the logic of the site	1. For the first textbox entry on the first_inputs page, there will be an if statement around the code where the entry is used ensuring that the user entered a number. If the user did not enter a number, a message will pop up telling them to enter a number. 2. For the second textbox entry on the first_inputs page, there will be an if statement around the code where the entry is used ensuring that the user entered a number. If the user did not enter a number, a message will pop up telling them to enter a number. 3. (Also 4 and 5 but I don't want to rewrite them): There will be an if statement for each possible selection in the first textbox on the second_inputs page. There will be 3-5 options of different goals they can have for their procedure, and each goal will lead to different recommendations for molecules to select in the checkboxes as well as different verbiage in the procedure.
Loop (minimum 1) This loop must iterate over the	This loop will iterate over the list of Structure dataclass instances, and for each instance, it will pick out the ratio int and do math with it to determine the

list/dict of dataclass instances in your State attribute. Description of this loop and why it is meaningful to the site	quantity of the staining molecule needed.

LLM Conversation

If you used an LLM in any part of this work, copy and paste every part of that conversation below. Otherwise, you may be given a zero for the assignment. Obviously, these words will not all be yours, and this is the exception to the requirement at the top of this assignment that all responses be in your own words. If you did not use an LLM to assist you in this assignment, simply say so below.

I did not use an LLM for any part of this work.

Final Submission

When you are finished, download this file as a Word Document (docx) and upload the file to the appropriate assignment on Canvas.